

Stefan Bobrowski

Frontend Engineer

Jacksonville, AR - stefanbobrowski1@gmail.com - 203-395-2674
<https://stefanbobrowski.com> - <https://www.linkedin.com/in/stefanbobrowski/>

Professional Summary

Frontend Engineer with 11+ years of professional experience delivering scalable web applications across high-traffic e-commerce websites, media platforms, and enterprise tools. Core expertise includes JavaScript, React, TypeScript, Vite, SCSS, Node.js, and Express, with a focus on UI/UX, performance, and accessibility. Currently pursuing Google Associate Cloud Engineer certification.

Technical Skills

Frontend: JavaScript, TypeScript, React, Vite, Next.js, Astro, Zustand, TanStack Query, SCSS Modules, Responsive Design, WCAG Accessibility

Backend: Node.js, Fastify, Express, REST, GraphQL, WebSockets, PostgreSQL, Redis, Upstash

AI Tools & APIs: GitHub Copilot, Claude, Gemini, Vertex AI, OpenAI

Cloud: Google Cloud Platform, Cloud Build, Cloud Run, Cloud Storage, Cloud SQL, BigQuery, Firebase

DevOps: Git, GitHub, GitHub Actions, Docker, CI/CD, Vitest, React Testing Library, Terraform

Security: OAuth 2.0, Rate Limiting, reCAPTCHA v3, JWT, IAM, WIF, Helmet, CORS

Analytics & SEO: Google Analytics 4, Google Search Console, Core Web Vitals, React Helmet

Visualization: Three.js, D3.js, React Three Fiber, Chart.js, Canvas, WebGL

Design: Figma, Adobe Photoshop, Adobe Premiere Pro

Professional Experience

Frontend Developer, Prepaire Labs (Contract)

July 2025 - September 2025 | Remote

- Architected a React + TypeScript web application using Vite for designing targeted therapeutics. Users could explore diseases, molecular targets, antibodies, and drug candidates through an interactive workflow. Delivered under a tight deadline that supported a successful seed funding demo.
- Integrated MolStar 2D/3D molecular visualization into React application, enabling users to dynamically render and explore molecular structures based on selected targets and compounds.
- Collaborated with UX lead in Figma to refine interaction flows, animations, and component structure.
- Implemented CI/CD pipeline using GitHub Actions and Docker, deploying to Microsoft Azure App Service.
- Contributed to full-stack architecture decisions including cloud platform selection, API design, and backend integration strategy.

Frontend Developer, Camelot SI LLC, SharperImage.com

Sep 2023 - Feb 2025 | Remote

- Implemented CRO-focused UI/UX redesigns across critical e-commerce flows (checkout, cart, login, returns, PIP/PDP) within a Next.js and TypeScript architecture serving thousands of users daily.

- Contributed to Bitbucket Pipelines CI/CD workflow across dev, staging, and production; regularly merged PRs, cut releases, and executed early-morning deployments and hotfixes.
- Integrated feature flags and A/B tests via Contentful CMS to support dynamic marketing campaigns and business experiments.
- Reduced frontend bundle size from ~50MB to ~12MB by auditing dependencies, removing unused packages, and optimizing build configuration, resulting in a ~70% build time reduction (10 min to 3 min).
- Consolidated multiple GraphQL requests into a single batched query, reducing API overhead and improving frontend performance.
- Delivered features and bug fixes for the internal dashboard used by warehouse and customer support teams; implemented Git Flow branching strategy and documented deployment process in Confluence.

Software Engineer, The Home Depot

May 2021 - Dec 2022 | Remote

- Led development of the Questions & Answers component seen on all PDP pages (React, GraphQL), directly improving CTR and user engagement.
- Served as Release Captain, reviewed pull requests, wrote release notes in Confluence, and deployed releases to production in a custom CI/CD pipeline.
- Contributed to a Core Web Vitals initiative by developing a placeholder system for SSR-hydrated elements, reducing Cumulative Layout Shift (CLS) on product detail pages from 0.50+ to 0.16.
- Increased test coverage to 100% for several components by transitioning from Enzyme to RTL + Jest.

Software Developer, Vibrant Software

July 2020 - May 2021 | Remote

- Architected and led development of the Nomad Portal, a digital asset management platform built with React, Redux-Saga, and an Express backend API.
- Integrated AWS AI services for content facial recognition, automated metadata tagging, and advanced search across rich media.
- Built VEP Virtual Stage, a live-streaming platform deployed for live concerts with hundreds of concurrent viewers. Features a multi-stream viewing grid, live chat via WebSockets, donation form workflow, and fully configurable content and theme using JSON.

Full-Stack Web Developer, Hat Trick Designs LLC

June 2019 - July 2020 | Remote

- Developed full-stack apps with MongoDB and the MERN stack.
- Converted React Class Components to Hooks and implemented state management with useContext.
- Backend REST API development with Node.js, Express, and Mongoose.

Frontend Web Developer, Direct Wines

February 2019 - May 2019 | Norwalk, CT

- Developed product microsites from designs and a customer checkout component for payment API requests using vanilla JavaScript, fetch, and async/await.
- Improved SEO and accessibility with metadata and Structured Data markup, allowing for search results to display custom widgets such as reviews and featured links.
- Researched and tested emerging frontend technologies such as Web Components, Vue, and React with Hooks for use in future code base.

Frontend Web Developer, Kathy Kuo Home

December 2017 - September 2018 | Harlem, NY

- Shipped new features in an ASP.NET MVC environment, weighed UI/UX decisions with Design Lead and CEO.
- Created sister-site and business portfolio <https://www.kathykuohome.com/designbar> from the ground up.
- Implemented Google Analytics and Google Optimize A/B tests, correlated increased CTR for new designs.
- Increased Google PageSpeed Insights score by 300% and reduced average page load time by 3 seconds.

Frontend Web Developer, Archi FX LLC, Healthcasts, Finalsight, oDesk

September 2014 - December 2017

- Archi FX LLC (Remote): Led frontend development on responsive web designs with flexbox.
- Healthcasts (New York, NY): Implemented product microsites containing sensitive medical information.
- Finalsight (Glastonbury, CT): Built responsive websites for schools globally in a fast-paced agile environment.
- oDesk (Remote): Completed various entry-level freelance jobs using HTML, CSS, JavaScript, and jQuery.

Education

Bachelor of Science in Computer Science, University of Connecticut

September 2010 - May 2014 | Storrs, Connecticut

Professional Development

September 2025 - Current

- Pursuing Google Cloud Platform (GCP) Associate Cloud Engineer certification.
- Building full-stack cloud-native projects and experimenting with cloud services.
- Integrating AI into applications using OpenAI, VertexAI, and Gemini APIs.